

生态研究前沿速览_2026-05-09

生态研究前沿速览

2026-05-09

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海洋与水生

1. Crop-climate models need more realistic representations of irrigation under warming (作物-气候模型需要更真实地表征变暖下的灌溉作用)

Nature Water | 2026-05-05 | 【Latest】

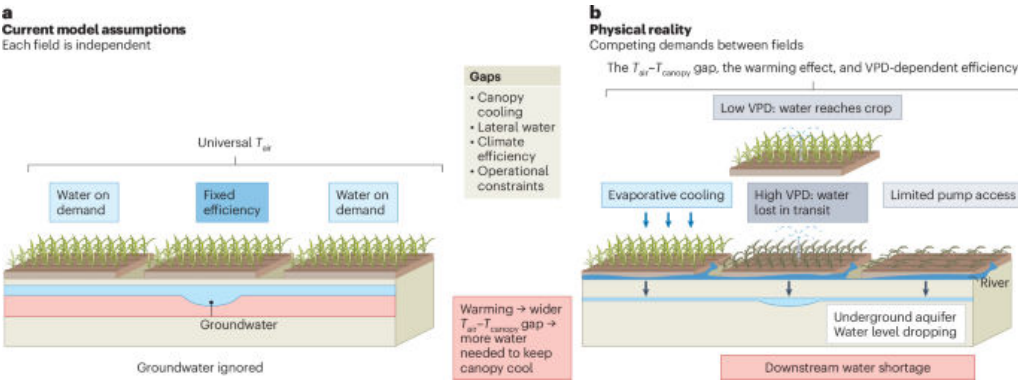


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Water, Published online: 05 May 2026; doi:10.1038/s44221-026-00642-9 Crop models increasingly project that irrigation can offset agricultural losses from climate change, informing major adaptation investments worldwide. Yet these models may systematically overestimate irrigation's protective capacity because the representation of how irrigation alters crop thermal environments remains incomplete in most large-scale assessments and assumes static efficiency under changing atmospheric conditions. More realistic assessments are urgently needed to avoid underestimating future water demands and regional adaptation failures.

中文摘要

作物模型越来越多地预测灌溉能够抵消气候变化造成的农业损失，并据此影响全球适应投资。但该研究指出，多数大尺度作物-气候模型对灌溉如何改变作物热环境的刻画仍不充分，且常假设灌溉效率在变暖大气条件下保持不变，因此可能系统性高估灌溉的保护能力，低估未来需水量和区域适应失败风险。

深度解读

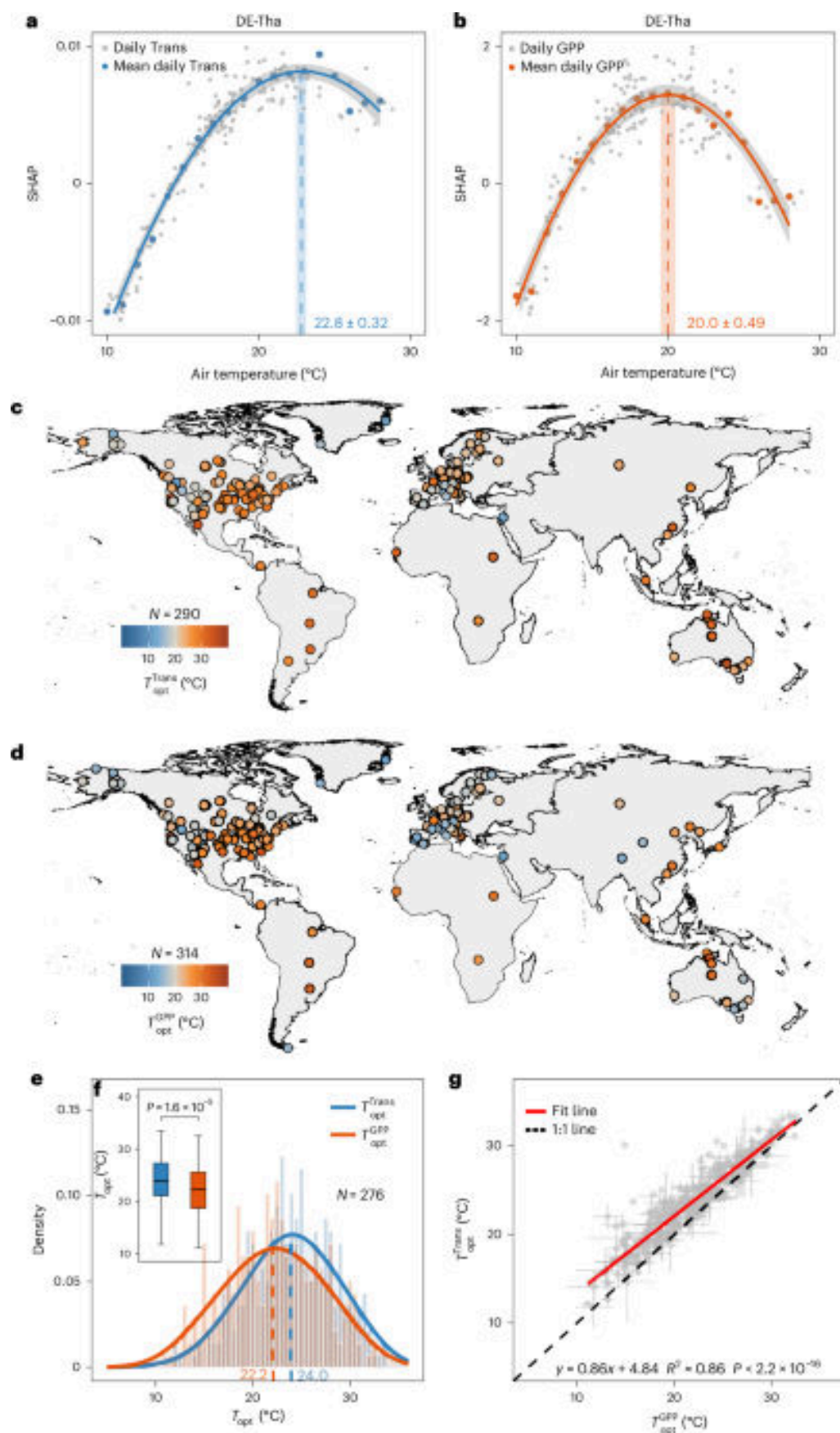
这篇评论的价值在于指出气候适应评估中的关键模型偏差：灌溉不是一个静态、无限可扩展的保护措施。变暖会改变蒸散需求、冠层温度和水资源竞争，使灌溉收益具有明显区域边界。若模型继续高估灌溉缓冲效应，农业适应投资可能低估水资源压力。对生态环境管理而言，粮食安全、地下水消耗和流域生态流量必须被联合评估。

DOI: <https://doi.org/10.1038/s44221-026-00642-9>

综合顶刊

2. Historical legacies shape continental variation in contemporary mammal food webs (历史遗产塑造当代哺乳动物食物网的大陆差异)

PNAS | 2026-04-27 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

SignificanceTrophic interactions underpin ecosystem structure and function, yet how past species losses, evolutionary histories, and environmental variability shape macroecological patterns of food webs remains poorly understood. We present continental-

中文摘要

研究指出，营养相互作用支撑生态系统结构与功能，但过去物种丧失、演化历史和环境变异如何共同塑造大陆尺度食物网格局仍缺乏清晰认识。作者从大陆尺度比较当代哺乳动物食物网，强调历史遗产会影响今天不同区域食物网的组成、连接方式和功能结构。

深度解读

这篇论文的重要性在于把历史灭绝和当代食物网结构联系起来，避免把当前生态网络视为静态或完全由现时环境决定。大型哺乳动物和顶级消费者的历史丧失会改变营养级结构，并影响生态系统功能恢复的上限。对保护规划而言，恢复食物网需要考虑区域历史基线，而不仅是现存物种清单。该研究也为再野化和大型动物恢复提供了更宏观的比较框架。

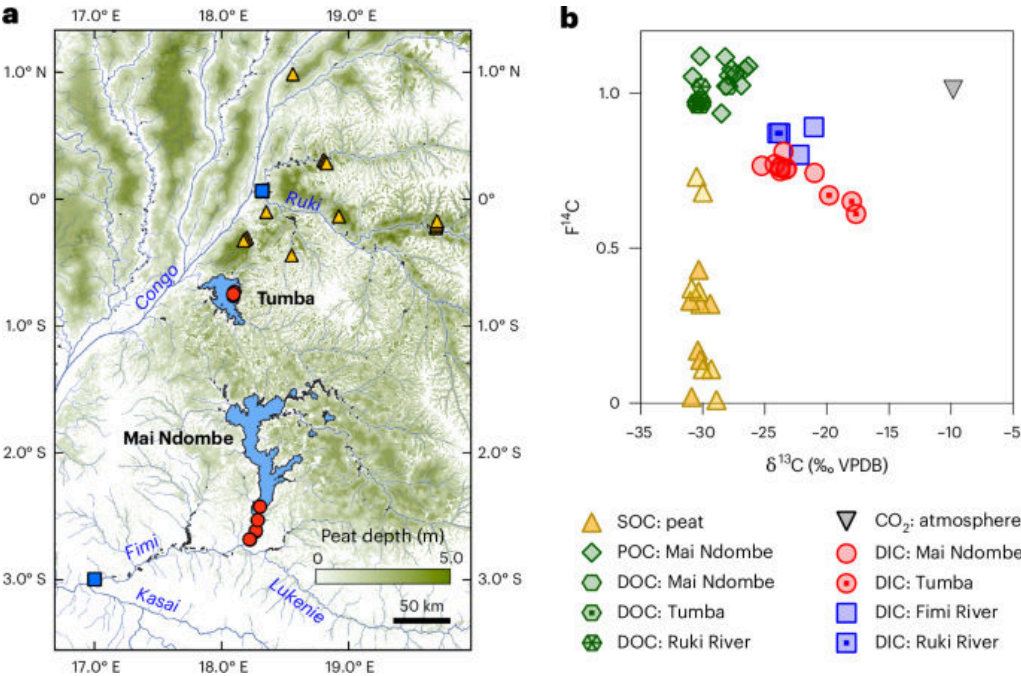
DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2519938123>

植物与农业

3. Habitat-Mediated Filtering, Rather Than Soil Properties, Shapes Plant Community Diversity on Nelson Island, Maritime Antarctica

(栖息地介导的过滤而非土壤性质塑造南极海洋性纳尔逊岛植物群落多样性)

Journal of Vegetation Science | 2026-05-09 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Journal of Vegetation Science, Volume 37, Issue 3, May/June 2026.

中文摘要

研究以海洋性南极纳尔逊岛植物群落为对象，比较栖息地条件和土壤性质对群落多样性的作用。结果表明，决定当地植物群落多样性和组成的主要因素是栖息地介导的环境过滤，而不是单一土壤理化属性，强调在极地生态系统中微地形、暴露度和生境结构对植物分布具有关键影响。

深度解读

极地植物群落物种少、环境梯度强，因此很适合识别群落装配机制。该文强调栖息地过滤优先于土壤性质，提示气候变暖造成的积雪、冻融和裸地暴露变化可能快速重塑植物多样性。对南极生态监测而言，样地选择不能只看土壤指标，还应记录微生境结构。它也为预测极地植被扩张提供了更细的环境解释框架。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70145>

海洋与水生

4. Chemoproteomics reveals global occurrence of a phytoplankton lyase capable of reconverting progestogens (化学蛋白质组学揭示可再转化孕激素的浮游植物裂解酶具有全球分布)

Nature Water | 2026-05-05 | 【Latest】

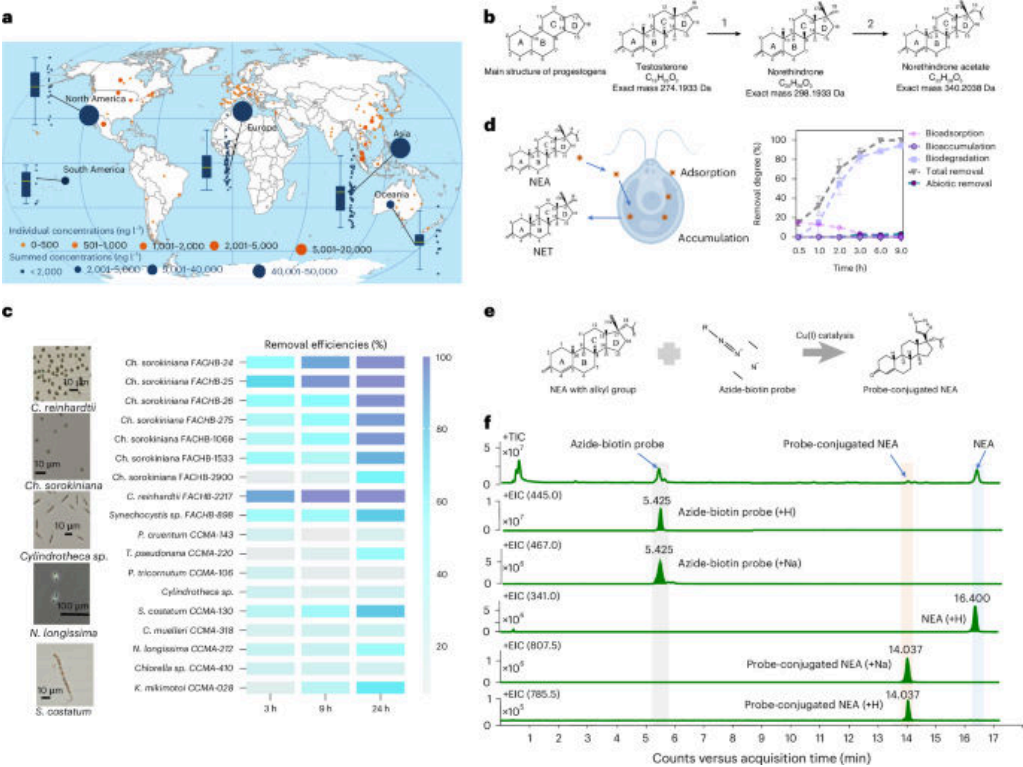


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Water, Published online: 05 May 2026; doi:10.1038/s44221-026-00646-5Endocrine-disrupting progesterogens pose risks to ecosystems through their biological activity. A chemoproteomics study shows that many phytoplankton species can transform these chemicals into forms that may be even more biologically active, potentially increasing ecological risk.

中文摘要

研究关注具有内分泌干扰风险的孕激素污染物。化学蛋白质组学结果显示，许多浮游植物物种携带一种裂解酶，能够把某些转化后的孕激素重新转化为可能更具生物活性的形式，说明水体中污染物的生态风险会受到浮游植物代谢网络调控，并可能在全球范围内发生。

深度解读

该研究把污染物风险从“降解即降低毒性”的简单假设中拉出来，显示生物转化可能使污染物重新获得活性。浮游植物处在水生食物网底层，其代谢过程会影响污染物暴露路径和生态效应。对水质管理来说，常规化学浓度监测可能不足以判断真实风险。未来需要把生物转化产物和关键酶的分布纳入新兴污染物评估。

DOI: <https://doi.org/10.1038/s44221-026-00646-5>

海洋与水生

5. Urea runoff fuels eutrophication of shallow surface waters in agricultural regions

(尿素径流推动农业区浅层地表水富营养化)

Nature Water | 2026-05-06 | 【Latest】

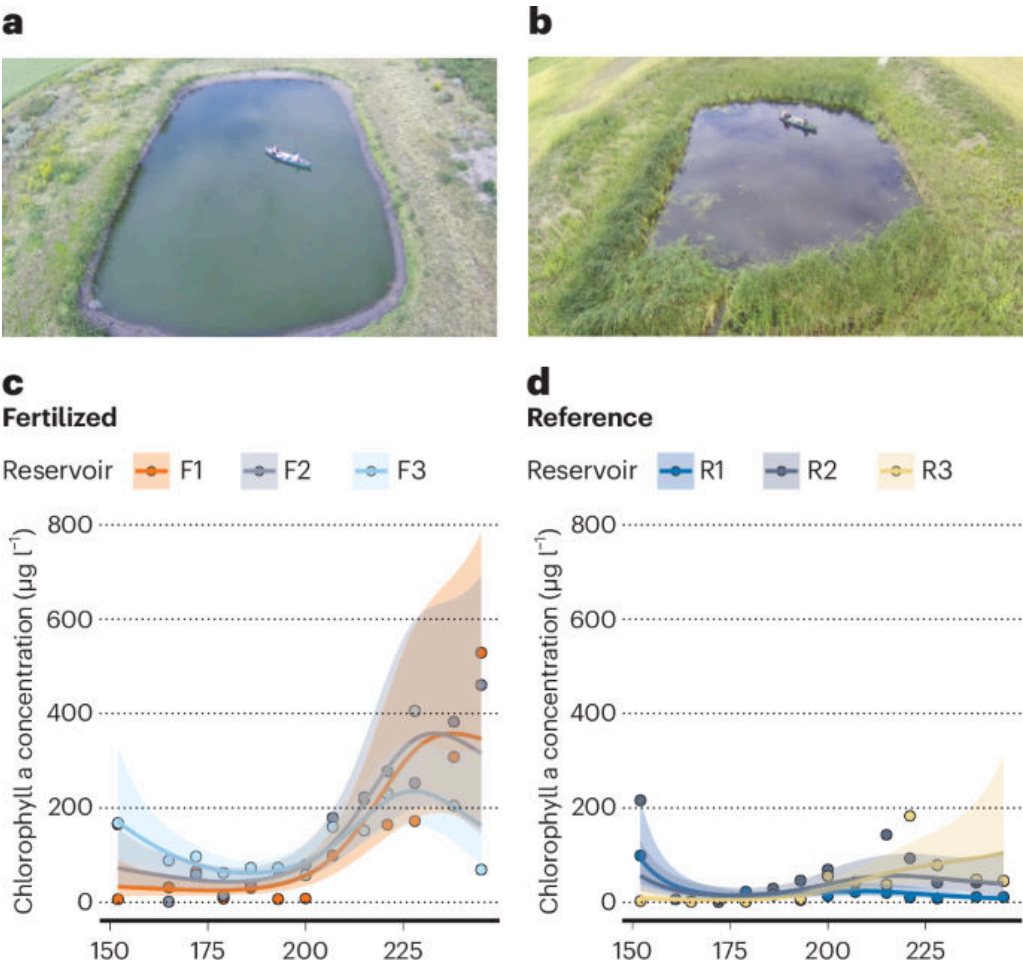


图 : Graphical abstract / article figure source: <https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2F>

Abstract

Nature Water, Published online: 06 May 2026; doi:10.1038/s44221-026-00641-w Urea is the world's most widely used nitrogen fertilizer. Elevated urea use can induce detrimental eutrophication of shallow surface waters in agricultural regions, accelerating water quality deterioration globally in these regions.

中文摘要

尿素是全球使用最广泛的氮肥。研究指出，农业区尿素使用增加会通过径流进入浅层地表水，促进有害富营养化过程，并加速这些区域的水质恶化。该结果把农业氮管理与湖泊、沟渠和浅水湿地生态风险直接联系起来。

深度解读

尿素常被视为农业生产资料，但进入浅水系统后会成为富营养化的直接燃料。浅层水体交换弱、光照充足，对外源氮输入特别敏感，因此风险可能比大河湖更快显现。该研究提示农业面源污染治理需要区分不同氮形态，而不仅计算总氮负荷。对流域管理来说，施肥时序、缓冲带和农田排水系统都应纳入控制策略。

DOI: <https://doi.org/10.1038/s44221-026-00641-w>

植物与农业

6. Turning Mediterranean Farmlands Into Priority Habitats: Natural Expansion of Juniper Woodlands After Agricultural Abandonment

(农业废弃后刺柏林自然扩张将地中海农田转变为优先栖息地)

Journal of Vegetation Science | 2026-05-02 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Journal of Vegetation Science, Volume 37, Issue 3, May/June 2026.

中文摘要

研究关注地中海农田废弃后刺柏林自然扩张的生态后果，指出部分废弃农田可逐步转化为具有保护价值的优先栖息地。该结果说明，土地利用退出并不总是退化过程，在合适区域也可能促进半自然木本群落恢复。

深度解读

地中海地区的农业废弃常被视为社会经济问题，但生态结果取决于恢复路径和目标物种。刺柏林自然扩张显示，被动恢复在某些景观中可以形成有保护价值的栖息地。该研究对欧洲和半干旱地区土地利用转型具有启发：不是所有废弃地都需要高强度造林或工程修复。关键是识别哪些地块适合自然恢复，哪些地块需要主动管理以避免火险或单一化。

综合顶刊

7. Historic logbooks reveal spatial footprints of commercial whaling (历史航海日志揭示商业捕鲸的空间足迹)

PNAS | 2026-04-27 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

SignificanceCommercial whaling devastated bowhead whale populations, and recovery has been slow and spatially heterogeneous. By reconstructing daily routes and hunting activities of more than 700 historic British and American whaling voyages, we mapped ...

中文摘要

研究利用 700 多次英美商业捕鲸航行的历史日志，重建捕鲸路线和猎捕活动的空间足迹。结果表明，商业捕鲸对弓头鲸等鲸类种群造成深刻影响，且恢复过程存在明显空间异质性，历史捕捞压力需要纳入当代海洋保护评估。

深度解读

历史生态数据常被低估，但它们能揭示现代监测开始前已经发生的强烈人类影响。商业捕鲸的空间足迹不仅解释了鲸类恢复不均，也会影响海洋碳循环、营养盐再分配和食物网结构。该研究提示海洋保护目标应纳入历史捕捞压力和空间记忆，避免把已经贫化的海洋状态当作自然基线。对生态修复而言，历史文献数据是重建长期基线的重要证据来源。

DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2523917123>

综合顶刊

8. Repeated snowball–hothouse cycles within the Neoproterozoic Sturtian glaciation (新元古代斯图尔特冰期内反复出现雪球-温室循环)

PNAS | 2026-04-27 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

SignificanceEarth experienced extreme climate swings during the Neoproterozoic epoch, including the Sturtian glaciation, when ice likely covered the planet. Explaining aspects of the geologic record and the survival of life through this event has been a ...

中文摘要

研究围绕新元古代斯图尔特冰期的极端气候波动展开，讨论冰封地球状态下雪球与温室条件可能反复交替。该结果有助于解释地质记录中的异常信号，也为理解极端气候转换期间生命如何持续提供背景。

深度解读

这项研究虽然偏古气候，但对生态环境速览有价值，因为它揭示地球系统可在极端状态之间发生反复跃迁。极端气候循环会改变风化、海洋化学和生物可利用栖息地，对生命演化背景具有根本影响。它提醒我们，气候系统存在非线性阈值和反馈，不能简单按线性外推理解长期生态风险。现代气候变化尺度不同，但阈值思维仍值得借鉴。

DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2525919123>

生态学核心

9. Increasing Mortality of Rare Tree Species Amplifies Extinction Risk in Tropical Forests Under Climate Change

(气候变化下稀有树种死亡率上升放大热带森林灭绝风险)

Global Ecology and Biogeography | 2026-05-06 | **【Latest】**



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究聚焦热带森林中稀有树种在气候变化下的灭绝风险。其核心观点是，稀有树种本就受低丰度和空间分布限制影响，当气候压力导致死亡率进一步上升时，种群更新失败和局地消失风险会被放大，从而削弱热带森林的物种多样性和长期稳定性。

深度解读

稀有树种往往对森林 beta 多样性和功能冗余具有不可替代作用，但常被总体生物量指标掩盖。该研究提醒，气候变化影响不只体现在平均生长下降，还会通过死亡率差异改变群落组成。若保护和恢复项目只关注常见优势树种，可能低估多样性丧失速度。热带森林监测需要更重视稀有种死亡、更新和空间连通性。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/geb.70235>

土壤与生物地球化学

10. Not just a carbon story: uncertainties related to the significance of microbially-produced nitrous oxide during soil carbon sequestration

(不只是碳故事：土壤固碳过程中微生物氧化亚氮重要性的不确定性)

Biogeochemistry | 2026-05-04 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

In the last decade, in extensive debates on the potential of soil carbon sequestration to decrease the amount of greenhouse gases (GHGs) emitted by soils, and thereby help mitigate climate change, discussions have centred overwhelmingly on CO₂. Separate research efforts have shown that other greenhouse gases, and in particular N₂O, produced by soil microorganisms, can have a sizeable impact on climate change. At this stage, it is unclear to what extent climate mitigation benefits due to increased soil carbon storage could be offset by N₂O emissions from soils. In the present article, we critically review the relevant primary literature. We conclude that reliable data do not exist at this point to determine whether soil carbon sequestration can help mitigate climate change at all. This is i

中文摘要

该综述指出，关于土壤固碳减缓气候变化的讨论长期集中于二氧化碳，但土壤微生物产生的氧化亚氮也可能对气候产生显著影响。现有证据尚不足以判断增加土壤碳储量带来的减排收益是否会被 N₂O 排放部分抵消，因此土壤固碳评估需要同时考虑多种温室气体。

深度解读

土壤固碳常被包装为自然气候解决方案，但温室气体收支必须算完整。N₂O 的增温潜势高，如果固碳管理同时增加反硝化或硝化相关排放，净气候收益会被削弱。该综述的重要性在于把土壤碳项目从单指标核算推向多气体、过程化评估。政策层面应避免只奖励土壤有机碳增加而忽略氮循环副作用。

DOI: <https://link.springer.com/article/10.1007/s10533-026-01330-2>

土壤与生物地球化学

11. Divergent biogeochemical responses in upland coastal forest soils to repeated flooding and shifts in water chemistry

(滨海高地森林土壤对反复淹水和水化学变化呈现分化的生物地球化学响应)

Biogeochemistry | 2026-05-07 | 【Latest】



图： Graphical abstract / article figure source:

Abstract

Sea level rise and storm surges drive hydrologic disturbances that alter biogeochemical processes in upland forests, contributing to forest mortality and ecosystem state change. However, the impact of repeated flooding on below-ground biogeochemistry in early stages of this transition is not well understood. We conducted a mesocosm flow-through incubation experiment, applying freshwater (FW) and brackish water (BW) pulses to intact soil monoliths (22 cm depth × 22 cm diameter) from a temperate upland coastal forest to understand how floodwater chemistry influences soil biogeochemistry and organo-mineral interactions. Continuous measurements of CO₂, CH₄, and N₂O fluxes were coupled with δ¹³C-CH₄ analysis, porewater chemistry (DOC, DIC, CDOM, SO₄²⁻, HS⁻, Fe²⁺, Mn²⁺, NH₄⁺, NO₃⁻ + NO₂⁻, Eh, pH)

中文摘要

海平面上升和风暴潮会改变滨海高地森林的水文扰动并推动生态系统状态转变。研究通过完整土柱的流通培养实验，比较淡水和半咸水反复脉冲对土壤 CO₂、CH₄、N₂O 通量、孔隙水化学和有机-矿物相互作用的影响，发现不同水化学条件会引发分化的地下生物地球化学响应。

深度解读

滨海森林向湿地或退化地转变往往先从地下过程开始。淡水和半咸水脉冲造成的反应不同，说明海平面上升风险不能只用淹水频率描述，还要考虑盐度和离子组成。该研究对预测“幽灵森林”、碳释放和甲烷/氧化亚氮排放很有意义。管理上，滨海缓冲带恢复需要把水文连通和水化学变化纳入同一框架。

DOI: <https://link.springer.com/article/10.1007/s10533-026-01340-0>

土壤与生物地球化学

12. Species-specific chlorine and bromine accumulation in temperate mushrooms: on the possible role of fungi in halogen cycling

(温带蘑菇中氯和溴的物种特异性积累及真菌在卤素循环中的潜在作用)

Biogeochemistry | 2026-05-06 | 【Latest】



图： Graphical abstract / article figure source:

Abstract

Chlorine (Cl) and bromine (Br) are important but understudied elements in terrestrial ecosystems, and their accumulation in mushrooms remains poorly characterized. We determined Cl and Br content in 126 mushroom samples collected from a temperate coniferous forest plantation with uniform environmental characteristics. The concentrations of both elements were also determined in a representative soil profile and organic topsoil layers across the studied site. The halogen concentrations in mushrooms ranged from 30 to 35,900 mg Cl kg⁻¹ and 0.18 to 45.9 mg Br kg⁻¹ (dry mass). The accumulation of Cl showed species- and genus-specific patterns, and the Cl concentrations were up to a hundred times higher than those in the topsoil. The concentrations of Br in mushrooms were several orders of magnit

中文摘要

研究测定温带针叶林人工林中 126 个蘑菇样品及土壤剖面的氯、溴含量。结果显示，蘑菇对氯和溴的积累具有明显物种和属特异性，部分氯浓度可比表层土高出上百倍，提示真菌可能是陆地生态系统卤素循环中被低估的生物参与者。

深度解读

卤素循环在陆地生态学中相对边缘，但会影响有机物分解、污染物转化和元素迁移。蘑菇显示出的强烈物种特异性说明，真菌功能群可能改变森林生态系统中氯、溴的生物有效性。该研究为把真菌纳入生物地球化学模型提供了证据。后续需要验证这些积累模式是否会沿食物网传递或影响土壤微生物过程。

DOI: <https://link.springer.com/article/10.1007/s10533-026-01337-9>

植物与农业

13. Crop Rotation Index for measuring agricultural land use intensity

(用于衡量农业土地利用强度的作物轮作指数)

Agronomy for Sustainable Development | 2026-04-30 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

Crop rotations constitute a crucial way of managing farms. Scientific methods have studied rotations through different approaches. These either focus on the design of rotations, the performance of individual rotations (from a productivity, environmental performance, and/or soil recovery point of view), or the spatial pattern in the landscape. Despite these approaches, there is no method available to investigate the sustainability of a rotation at a large scale based on consistently followed agronomic principles in everyday farming practices. The novel Crop Rotation Index is proposed to fill this gap. To compute the Crop Rotation Index on a parcel basis, a sequence of crops over 6 years is evaluated by 8 indicators, where crops in the sequence are grouped together based on botanical families.

中文摘要

研究提出作物轮作指数，用 6 年作物序列和 8 个农艺指标在地块尺度评估农业土地利用强度。该方法试图把轮作多样性、植物科属组合和管理原则转化为可在大尺度农业景观中比较的量化指标。

深度解读

轮作是连接产量、土壤健康、病虫害压力和生物多样性的关键管理措施，但大尺度量化一直困难。作物轮作指数把复杂的轮作序列压缩成可比较指标，为遥感、农情统计和可持续农业评估提供接口。虽然该条目评分较低，但它补齐了当前农业土地利用强度方法学方向。后续需要验证该指数是否能稳定预测土壤碳、农药使用和景观生物多样性结果。

DOI: <https://link.springer.com/article/10.1007/s13593-026-01101-1>

海洋与水生

14. From Lability to Conservatism: Evolutionary Patterns in the Traits of Stream Trichoptera

(从易变到保守：溪流毛翅目性状的演化格局)

Freshwater Biology | 2026-05-08 | 【Latest】



图：Graphical abstract / article figure source:

Abstract

中文摘要

研究关注溪流毛翅目昆虫性状从演化易变到保守的格局。通过比较性状在谱系中的变化，论文试图识别哪些生态功能性状容易随环境或谱系分化改变，哪些性状在演化上更稳定，为溪流昆虫功能多样性和淡水生态监测提供进化背景。

深度解读

淡水昆虫常被用于水质和栖息地评价，但指标解释若忽略演化历史，容易把谱系约束误判为环境响应。该研究通过性状演化格局帮助区分可塑性强的生态响应性状和较保守的系统发育性状。对生物监测来说，这能提升指标选择的机制性。虽然该条目分数较低，但它补充了淡水生态评价中功能性状和演化背景的交叉方向。

DOI: <https://onlinelibrary.wiley.com/doi/10.1111/fwb.70229>
